Unveiling Diabetes Risk Factors: Exploratory Data Analysis and Collinearity Decoding

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Introduction

This study performs an Exploratory Data Analysis (EDA) on the classic Pima Indians Diabetes Database (768 records). By identifying features like "Pregnancies" and a minimum age of 21, we confirmed the dataset exclusively tracks adult females. Initial observations revealed critical data quality issues—such as biologically impossible zero values in Insulin (48.7% missing)—highlighting the necessity of rigorous data cleaning prior to predictive modeling.

Objective

- To evaluate data integrity and identify missing or anomalous values.
- To discover the most significant standalone predictor for diabetes.
- To uncover hidden correlations among physiological features and resolve potential multicollinearity.

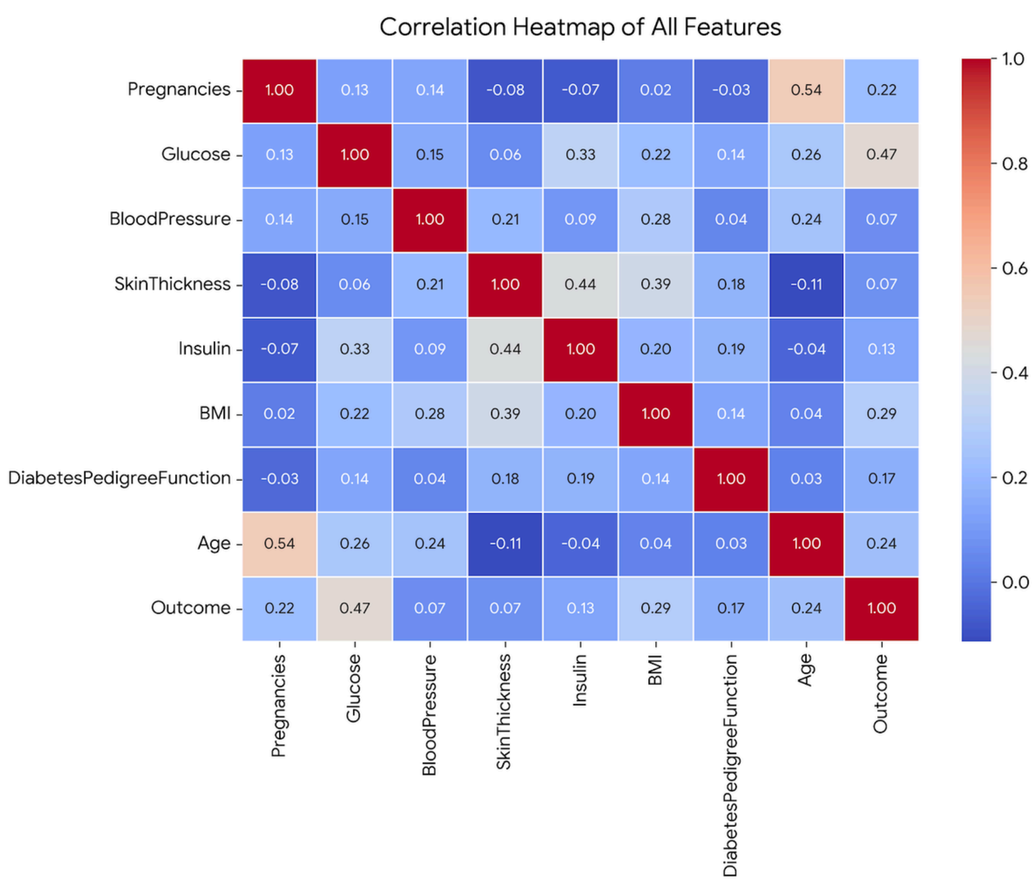
Methodology

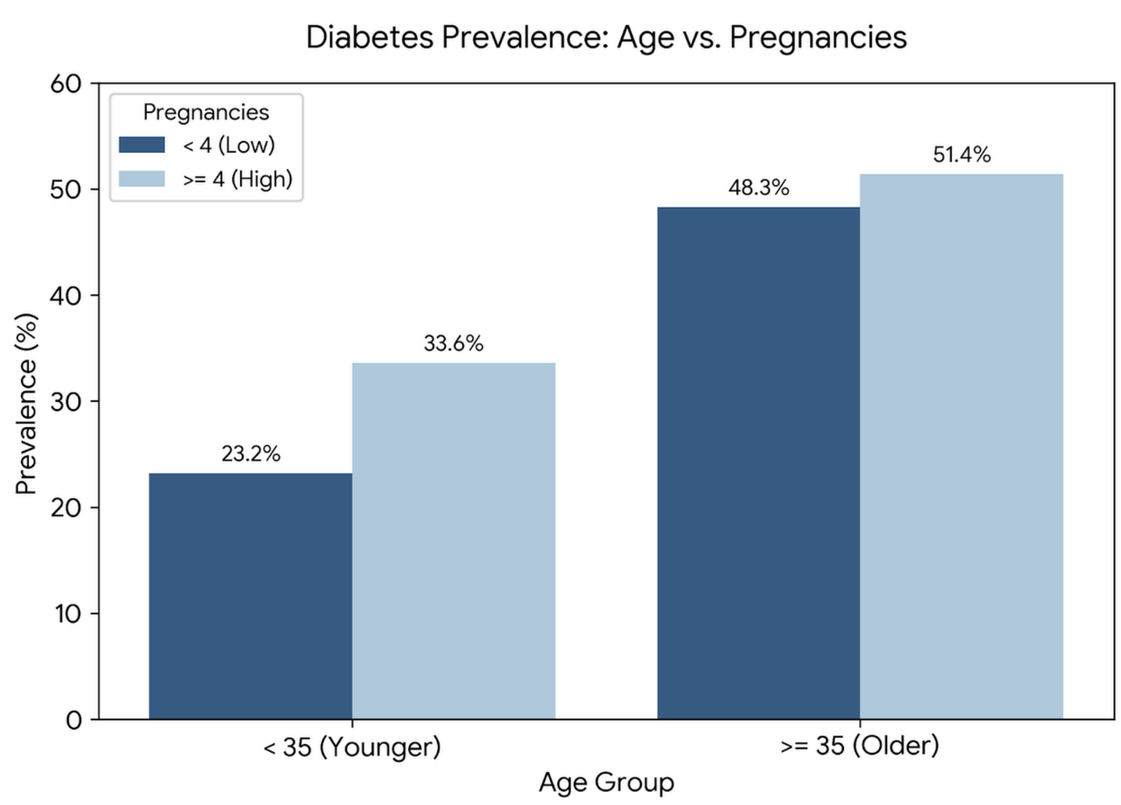
- Tools: Python for data cleaning and visualization.
- Statistical Validation: Conducted a 1,000-iteration Bootstrap sampling simulation (SEED=12) to empirically demonstrate the Central Limit Theorem (CLT) and ensure the statistical stability of sample means.

Results

Data quality checks revealed substantial zero-coded missing values across multiple metrics (e.g., Insulin 48.7%, Skin Thickness 29.5%). Among valid features, Glucose emerged as the most significant differentiator; the median glucose level and distribution range of the diabetic group are distinctly higher than those of the healthy group.

Analysis





Conclusion

This phase concludes with three final verdicts:

1. Data Quality Alert: Raw clinical data is inherently flawed. Variables with high missing rates (e.g., Skin Thickness) should be discarded or replaced by more robust metrics like BMI.
2. Glucose is King: Glucose remains the most reliable independent predictor.
3. Decoding Collinearity: The predictive weight of Age vastly exceeds that of Pregnancies. Future clinical risk models must prioritize "Advanced Age" over parity to avoid skewed assessments.

Related Literature

- Dataset: Pima Indians Diabetes Database (National Institute of Diabetes and Digestive and Kidney Diseases).
- Concepts: Central Limit Theorem (CLT), EDA, Correlation & Multicollinearity Analysis.